

Roll No.....

MID SEMESTER EXAMINATION SEP - 2025

Name of the Program: B.Tech Aerospace Engineering

Name of the Course: Aircraft Structure-I

Time: 90 minutes

Semester: Vth

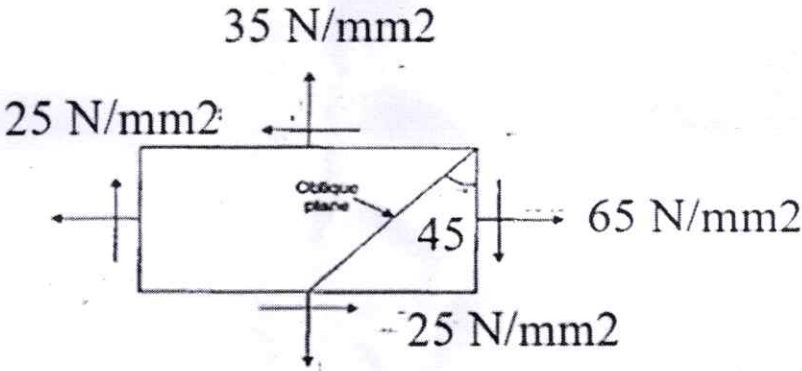
Course Code:TAS501

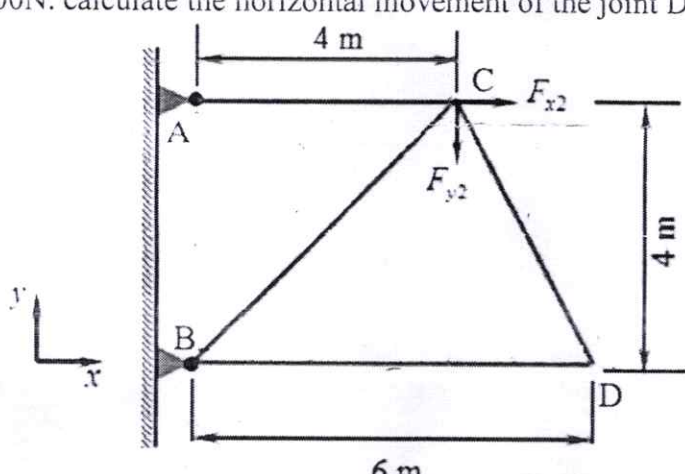
Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing *any one of the sub questions*.

(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Define the term stress and strain. Explain the different types of stresses and strains with suitable examples.	
OR		
(b)	At a particular point in a structural member a two-dimensional stress system exists where $\sigma_x = 60 \text{ N/mm}^2$, $\sigma_y = -40 \text{ N/mm}^2$ and $\tau_{xy} = 50 \text{ N/mm}^2$. If Young's modulus $E = 200\,000 \text{ N/mm}^2$ and Poisson's ratio $\nu = 0.3$. Calculate the direct strain in the x and y directions and the shear strain at the point.	
Q2	(10 marks)	CO1
(a)	Prove that the shear stresses are always complementary on perpendicular planes? ($\tau_{xy} = \tau_{yx}$, $\tau_{yz} = \tau_{zy}$ and $\tau_{zx} = \tau_{xz}$)	
OR		
(b)	What are compatibility equations? Derive the compatibility equation for any Two dimensional plane.	
Q3	(10 marks)	CO2
(a)	At a point in a strained material the principal stresses are 100 N/mm^2 (tensile) and 60 N/mm^2 (compressive). Determine the normal stress, shear stress and resultant stress on a plane inclined at 50° to the axis of major principal stress. Also determine the principal planes, principal stresses and the maximum shear stress.	
OR		
(b)	A point in a strained material is subjected to stresses shown in Fig below. Using Mohr's circle method, determine the normal and tangential stress across the oblique plane. 	
Q4	(10 marks)	CO2
(a)	Prove that the plane of maximum shear is at 45 degree from the principal planes?	
OR		

(b)	A large thin plate is subjected to certain boundary conditions on its thin edges (with its large faces free of stress), leading to the stress function $\phi = Ax^3y^2 - Bx^5$ a) use the biharmonic equation to express A in terms of B b) calculate all stress components c) check that the compatibility equation is satisfied d) check that the equilibrium equations are satisfied	
Q5	(10 marks)	
(a)	Consider the four-member truss as shown below. The members of the truss have a common cross-sectional area of 10 cm^2 and are made of the same material with elastic modulus $E = 110 \text{ GPa}$. The truss is subjected to $F_{x2} = 500 \text{ N}$ and $F_{y2} = 500 \text{ N}$. calculate the horizontal movement of the joint D. 	CO3
OR		
(b)	Determine the forces in each member of the pin jointed framework as shown below. Assume $A = 1000 \text{ mm}^2$ and $E = 210 \text{ GPa}$ same for all members. 